NLP

Natural Language Processing (NLP) is a field of AI that enables the computers to understand, interpret, generate and interact with human languages.

NLP combines linguistics, CS and ML a DL.

NLP powers technologies like ChatGPT, Siri, Alexa, Search Engines, Text Summarization etc.

Text Related Data → NLP
Photos, Images → Computer Vision
Time related data → Time Series

Text Preprocessing

1) Tokenization → Breaking a text into smaller units called tokens.

Eg I love machine learning → ['I', 'love', 'machine', 'learning']

2) Lemmatization → converts a word into its base dictionary form (lemma) considering grammar.

Eg 'studies' → 'study'
'better' → 'good'

3) Stemming → Removes word suffixes to reduce words to root form, but meaning may not stay intact.

Eg 'studies' → 'study'
'working' → 'work'

4) Stopwords Removal → Removes commonly used words that don't add meanings.

Eg 'the', 'is', 'and', 'was', 'in' etc.

Corpus → Corpus is simply a large and structured collection of texts used for language research or NLP.

Stopwords are the common words.

Text to Vector Techniques

→ After preprocessing, text must be converted into numbers for ML models :-

1) Bag of Words (BoW)

→ Counts frequency of each word in the text.

Eg 'I love ML', 'I love AI'

Vocabulary = {I, love, ML, AI}

Sentence	I	love	ML	AI
S_1	1	1	1	0
S_2	1	1	0	1

2) TF-IDF (Term Frequency, Inverse Document Frequency)

TF → How often a word appears in documen

IDF → How rare the word is across all documents.

TF-IDF = TF × IDF

Eg. Common words like 'is', 'the' get very low score.

Rare and meaningful words gets high score.

3) One Hot Encoding (OHE)

→ Represents each word as vector with 1s and 0s.

Eg ['cat', 'dog', 'rat']

Word	Vector
cat	[1,0,0]
dog	[0,1,0]
rat	[0,0,1]

Summary,

→ Tokenization → Break Text
→ Lemmatization → Convert to dictionary word
→ Stemming → Cut suffixes
→ Stopword Removal → Remove common words.

→ BoW → Word Counts
→ TF-IDF → rare words more important
→ OHE → Binary Vectors